# LAB CYCLE

Name: Neeraja
class:II BCA

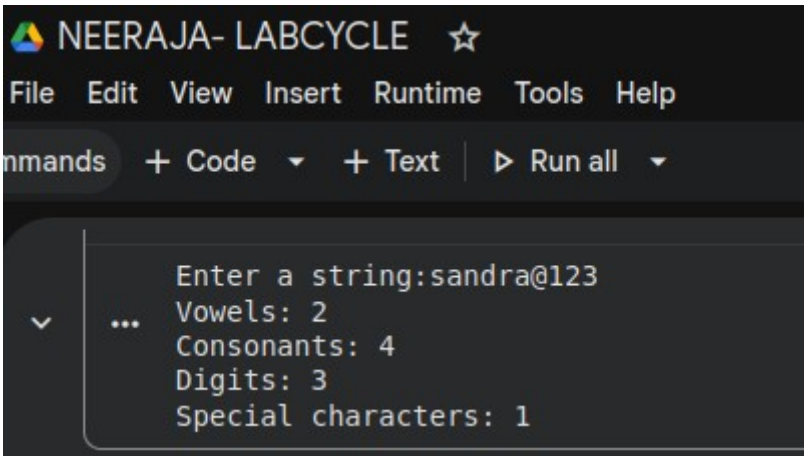
1. Write a program to count vowels, consonants, digits, and special characters in a string given as input.

Program:

```
string=input("Enter a string:")
vowels=0
consonants=0
digits=0
special=0
for ch in string:
    if ch.isalpha():
        if ch.lower() in "aeiou":
            vowels+=1
        else:
            consonants+=1
    elif ch.isdigit():
        digits+=1
    else:
        special+=1

print("Vowels:",vowels)
print("Consonants:",consonants)
print("Digits:",digits)
print("Special characters:",special)
```

Output:

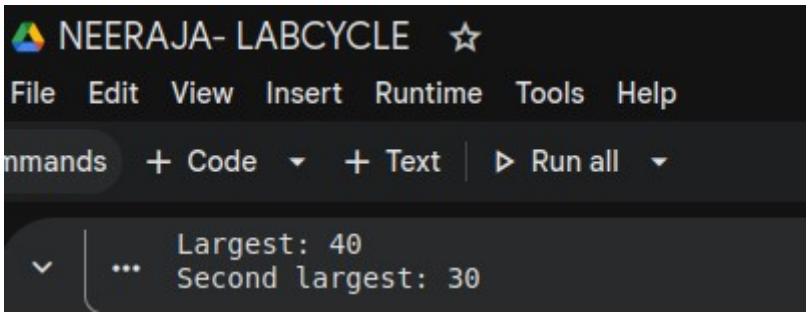


2.Write a program to find the largest and second largest element in a list.

Program:

```
largest=max(numbers)
second=max(x for x in numbers if x!=largest)
print("Largest:",largest)
print("Second largest:",second)
```

Output:



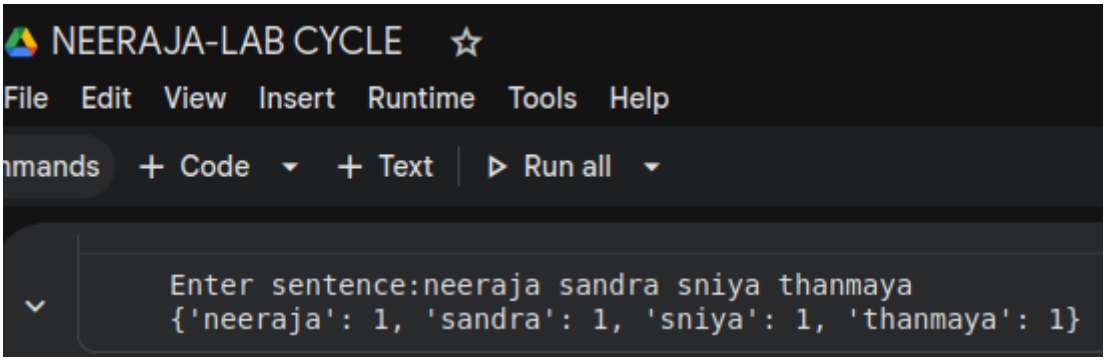
3.Write a program to count word frequency using a dictionary.

Program:

```
text=input("Enter sentence:")
frequency={}
word=""
for ch in text + " ":
    if ch!=" ":
        word+=ch
    else:
        frequency[word] = frequency.get(word, 0) + 1
        word=""

print(frequency)
```

Output:

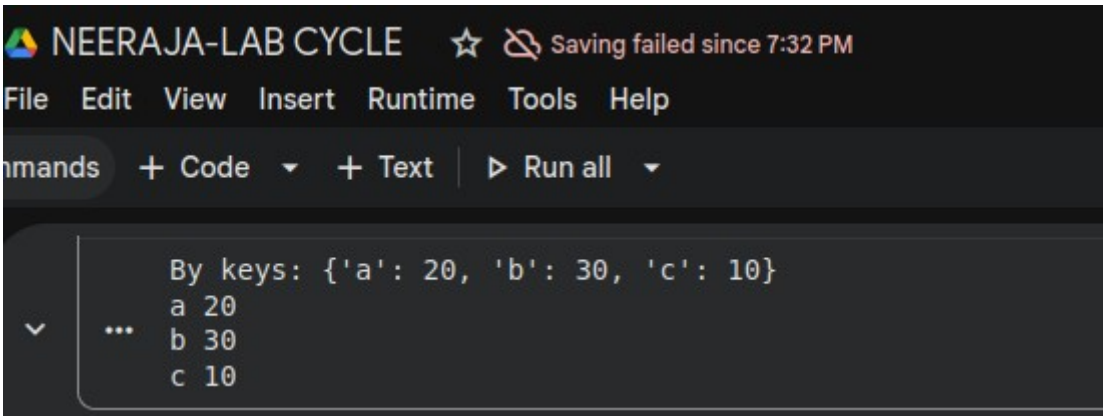


4.Write a program to sort dictionary by keys and the by values.

Program:

```
d={'b': 20, 'a': 10, 'c': 30}
print("By keys:", dict(sorted(d.items())))
for k, v in sorted(d.items()):
    print(k, v)
```

Output:

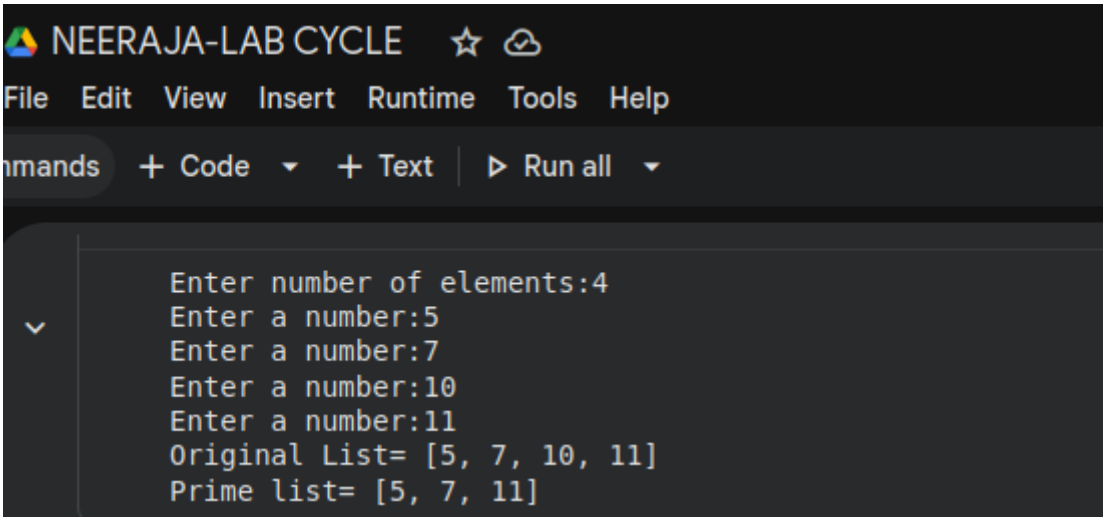


5.Write a program using list comprehension to generate prime numbers.

Program:

```
def check_prime(n):
  if n<=2:
    return True
  for i in range(2,(n//2)+1):
    if n%i==0:
      return False
    else:
      return True
l1=[]
a=int(input("Enter number of elements:"))
for i in range(a):
  x=int(input("Enter a number:"))
  l1.append(x)
print("Original List=",l1)
prime_of_l1=[x for x in l1 if check_prime(x)]
print("Prime list=",prime_of_l1)
```

Output:

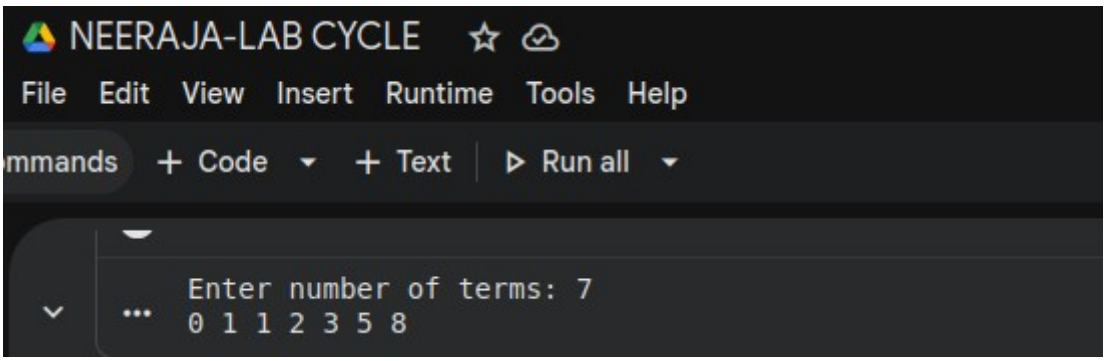


6.Write a program to generate Fibonacci series using loops.

Program:

```
n=int(input("Enter number of terms: "))
a,b= 0,1
for i in range(n):
    print(a,end=" ")
    a,b=b,a+b
```

Output:

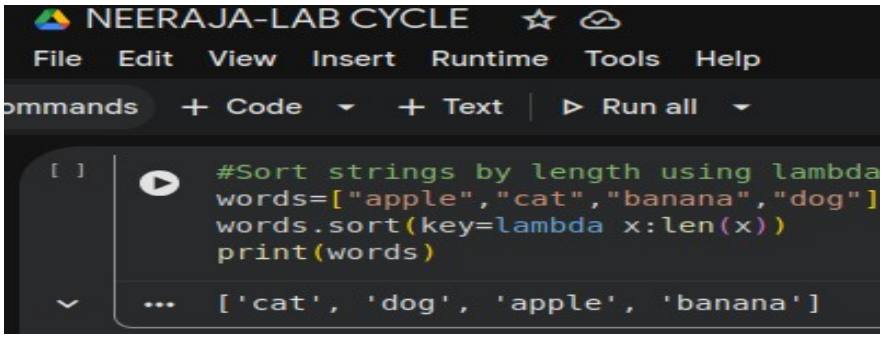


7.Sort strings by length using lambda function.

Program:

```
words=["apple","cat","banana","dog"]
words.sort(key=lambda x:len(x))
print(words)
```

Output:

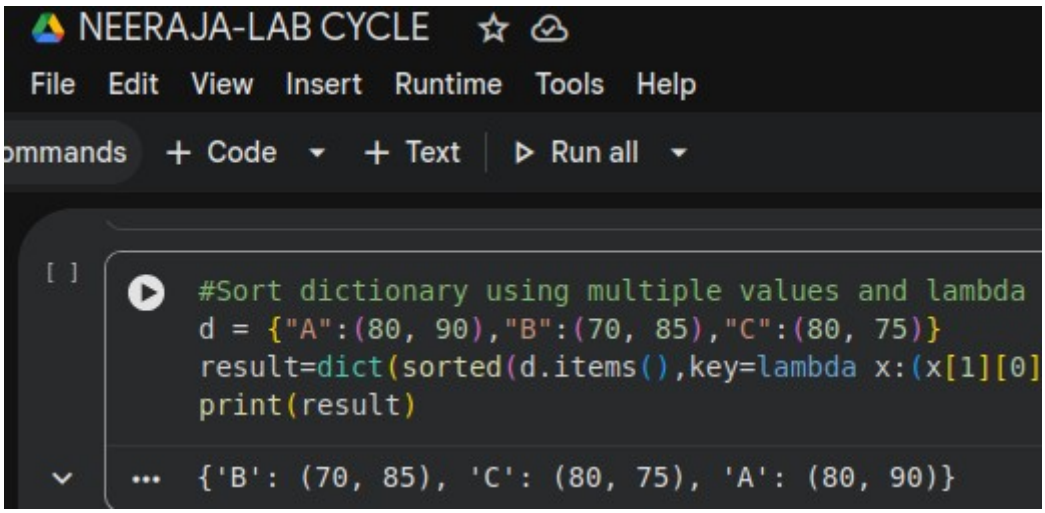


8.Sort dictionary using multiple values and lambda function.

Program:

```
d = {"A":(80, 90),"B":(70, 85),"C":(80, 75)}
result=dict(sorted(d.items(),key=lambda x:(x[1][0],x[1][1])))
print(result)
```

Output:



9.Write a program to demonstrate Inheritance and polymorphism.

Program:

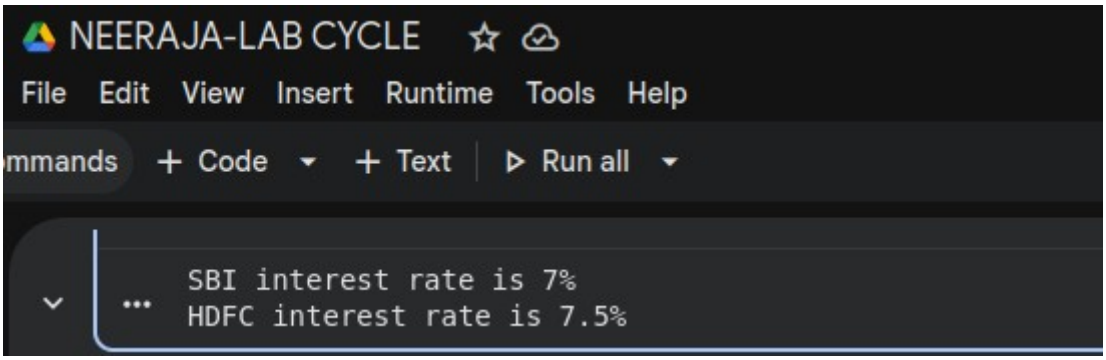
```
class Bank:
    def interest(self):
        print("Bank interest rate")

class SBI(Bank):
    def interest(self):
        print("SBI interest rate is 7%")


class HDFC(Bank):
    def interest(self):
        print("HDFC interest rate is 7.5%")

sbi = SBI()
hdfc = HDFC()
sbi.interest()
hdfc.interest()
```

Output:



10. Write a program to read and write text files .Use exceptions.

Program:

```
try:

    with open("sample.txt", "w") as file:
        file.write("Hello, World!\n")
        file.write("This is a Python file handling program.")

    print("Data written successfully.")


    with open("sample.txt", "r") as file:
        data = file.read()

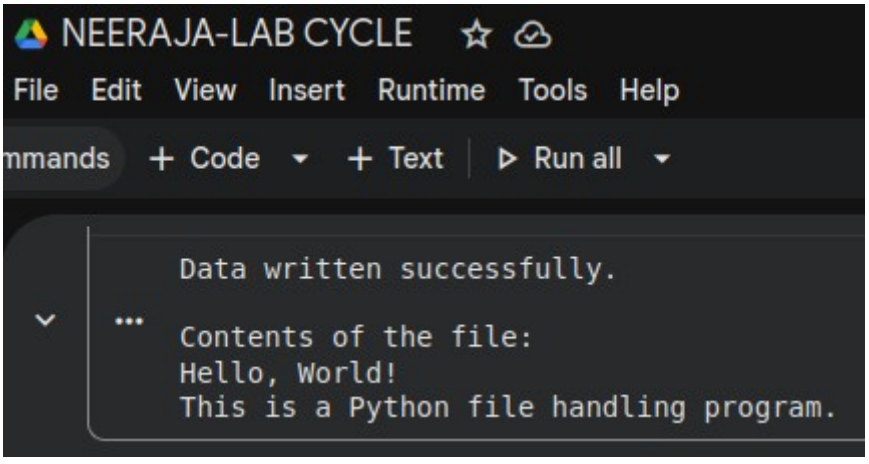
    print("\nContents of the file:")
    print(data)

except FileNotFoundError:
    print("Error: File not found.")

except PermissionError:
    print("Error: Permission denied.")

except Exception as e:
    print("An error occurred:", e)
```

Output:



11.Write a Python program using Pandas a. to create and analyze a student dataset containing student name, marks and grade. b. Use the following data: data = { "Name": ["Aman", "Diya", "Rahul", "Sneha", "Arjun"], "Marks": [78, 92, 85, 67, 74], "Grade": ["B", "A", "A", "C", "B"]} c. Perform the following operations:
i.Create a Pandas DataFrame using the given dataset.
ii.Display the complete dataset.
Iii.Display the first 3 rows of the dataset.
iv.Check whether the dataset contains missing values.
v.Create a histogram showing mark distribution.
vi.Find median and standard deviation of marks.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Name": ["Aman", "Diya", "Rahul", "Sneha", "Arjun"],
    "Marks": [78, 92, 85, 67, 74],
    "Grade": ["B", "A", "C", "B", "B"]
}
df = pd.DataFrame(data)

print("Complete Dataset:")
print(df)

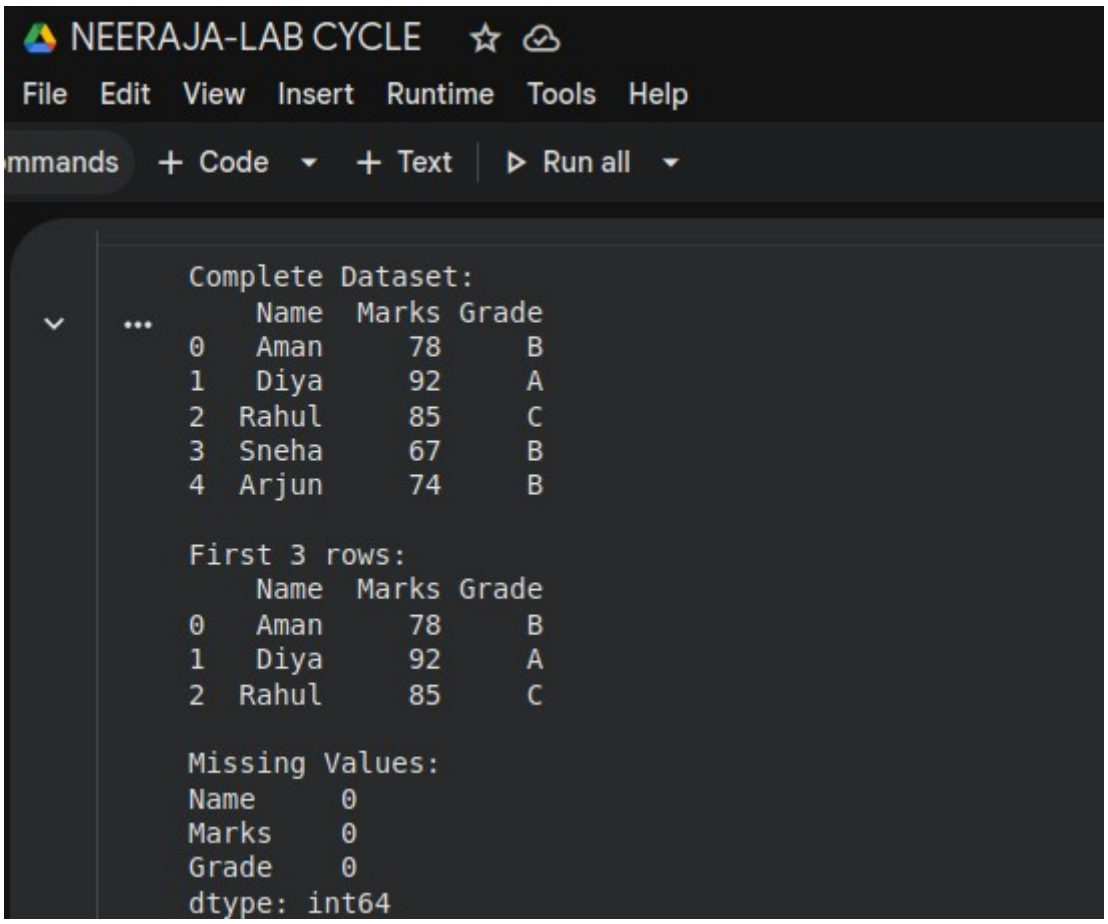
print("\nFirst 3 rows:")
print(df.head(3))

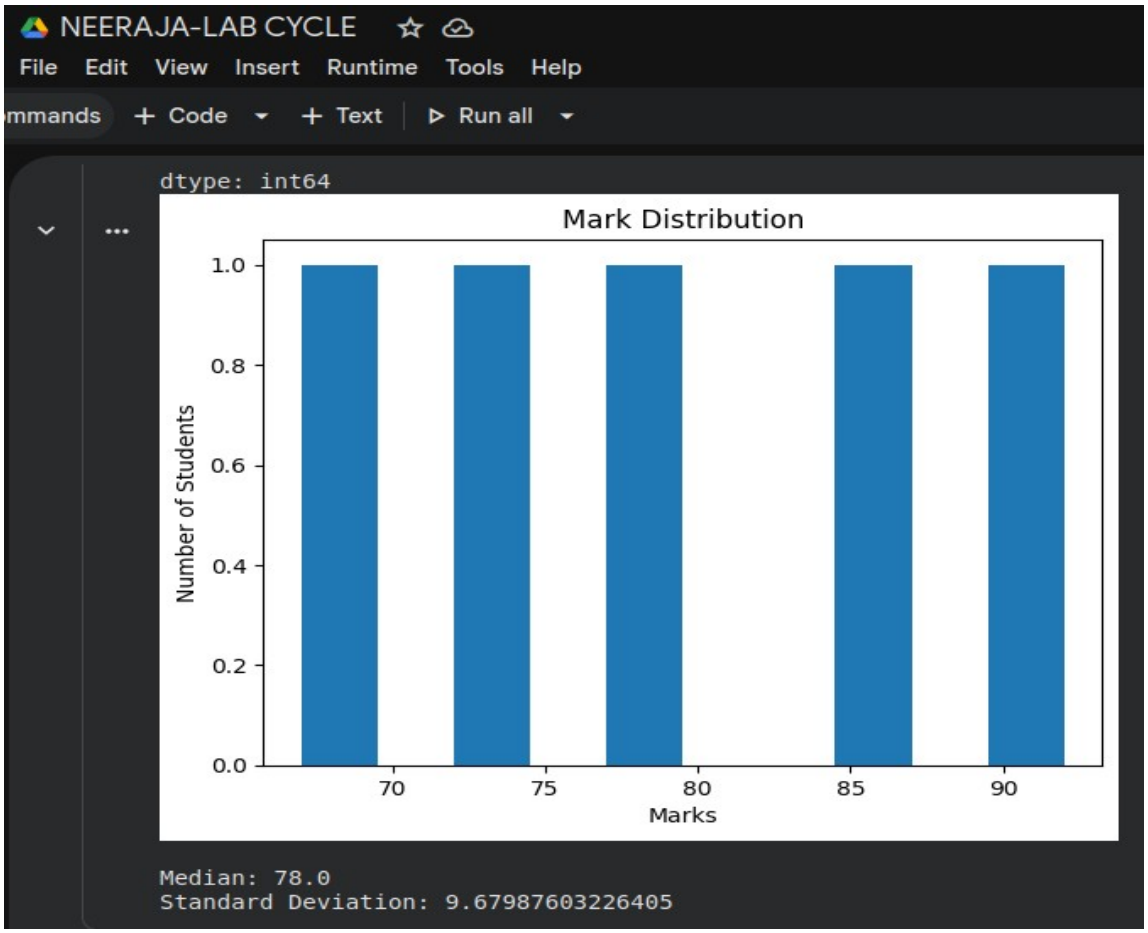
print("\nMissing Values:")
print(df.isnull().sum())

plt.hist(df["Marks"])
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Mark Distribution")
plt.show()

print("\nMedian:", df["Marks"].median())
print("Standard Deviation:", df["Marks"].std())
```

Output:


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File Edit View Insert Runtime Tools Help
mmands + Code + Text Run all
Complete Dataset:
Name Marks Grade
0 Aman 78 B
1 Diya 92 A
2 Rahul 85 C
3 Sneha 67 B
4 Arjun 74 B
First 3 rows:
Name Marks Grade
0 Aman 78 B
1 Diya 92 A
2 Rahul 85 C
Missing Values:
Name 0
Marks 0
Grade 0
dtype: int64


NEERAJA-LAB CYCLE
File Edit View Insert Runtime Tools Help
mmands + Code + Text Run all
dtype: int64
Mark Distribution
Number of Students
1.0
0.8
0.6
0.4
0.2
0.0
70
75
80
85
90
Marks
Median: 78.0
Standard Deviation: 9.67987603226405

12.Create dataframe from the given
a. data = {"Name": ["Aman", "Diya", "Rahul", "Sneha", Arjun"], "Department": ["CS", "IT", "CS", "ECE", "IT"], "Marks": [78, 92, 85, 67, 74], "Attendance": [85, 90, 88, 75, 80], "Grade": ["B", "A", "A", "C", "B"]}.
b. Display Dataset Information
c. Display Statistical Summary
d. Display only student names and marks
e. Display students scoring more than 80 marks.
f. Sort students based on marks in descending order.
g. Add a new column named “Bonus” containing: 5 marks bonus for every student.

Program:

```
import pandas as pd

data={"Name": ["Aman", "Diya", "Rahul", "Sneha", "Arjun"],
      "Department": ["CS", "IT", "CS", "ECE", "IT"],
      "Marks": [78, 92, 85, 67, 74],
      "Attendance": [85, 90, 88, 75, 80],
      "Grade": ["B", "A", "A", "C", "B"]}

df = pd.DataFrame(data)
print("Dataset Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nNames and Marks:")
print(df[["Name","Marks"]])

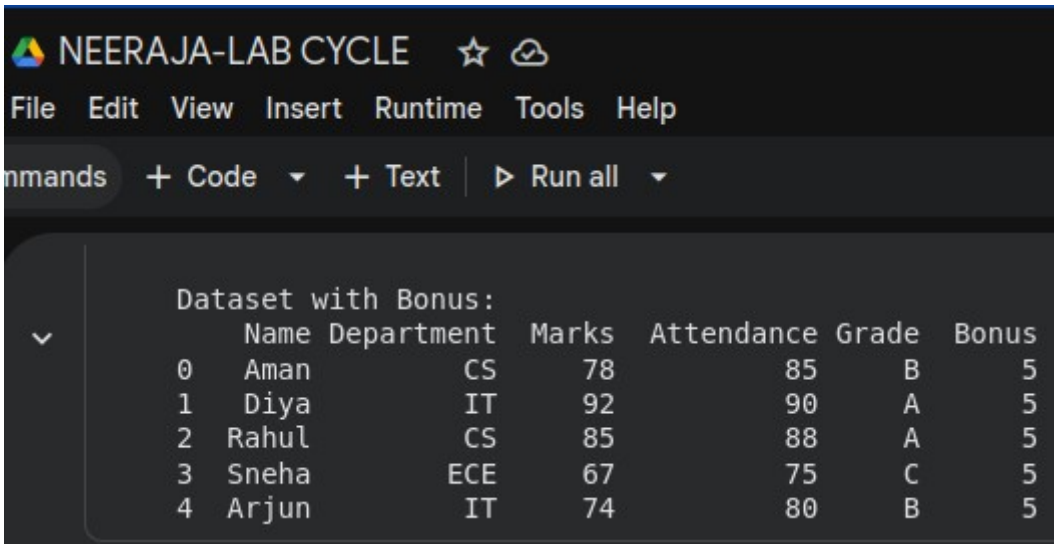
print("\nStudents scoring more than 80:")
print(df[df["Marks"]>80])

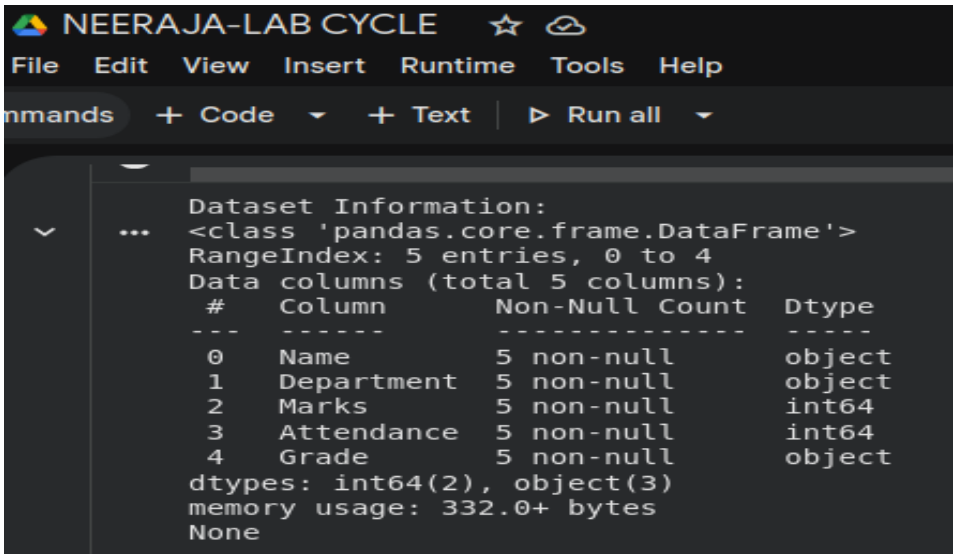
print("\nSorted by Marks:")
print(df.sort_values("Marks",ascending=False))

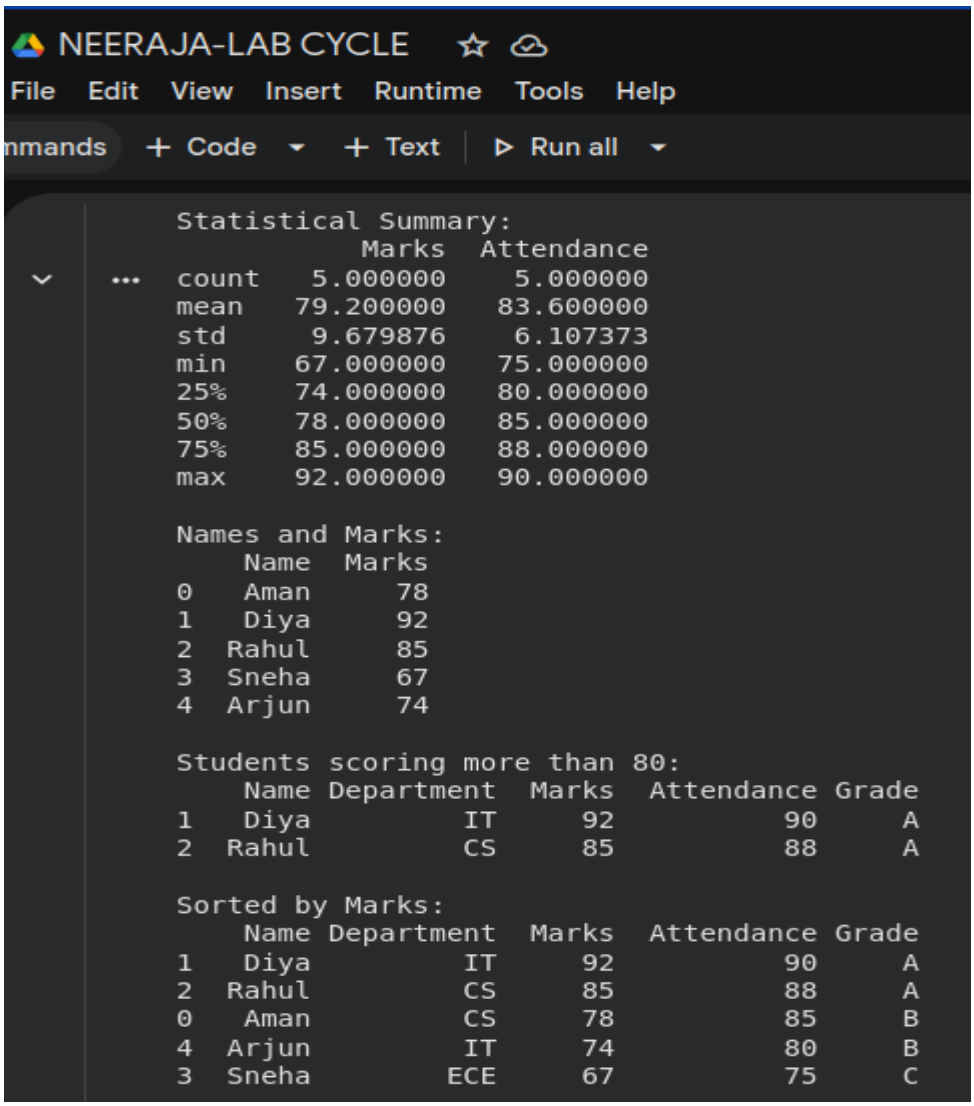
df["Bonus"]=5

print("\nDataset with Bonus:")
print(df)
```

Output:


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File Edit View Insert Runtime Tools Help
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Dataset with Bonus:
Name Department Marks Attendance Grade Bonus
0 Aman CS 78 85 B 5
1 Diya IT 92 90 A 5
2 Rahul CS 85 88 A 5
3 Sneha ECE 67 75 C 5
4 Arjun IT 74 80 B 5


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File Edit View Insert Runtime Tools Help
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Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
# Column Non-Null Count Dtype
0 Name 5 non-null object
1 Department 5 non-null object
2 Marks 5 non-null int64
3 Attendance 5 non-null int64
4 Grade 5 non-null object
dtypes: int64(2), object(3)
memory usage: 332.0+ bytes
None


NEERAJA-LAB CYCLE
File Edit View Insert Runtime Tools Help
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Statistical Summary:
Marks Attendance
count 5.000000 5.000000
mean 79.200000 83.600000
std 9.679876 6.107373
min 67.000000 75.000000
25% 74.000000 80.000000
50% 78.000000 85.000000
75% 85.000000 88.000000
max 92.000000 90.000000
Names and Marks:
Name Marks
0 Aman 78
1 Diya 92
2 Rahul 85
3 Sneha 67
4 Arjun 74
Students scoring more than 80:
Name Department Marks Attendance Grade
1 Diya IT 92 90 A
2 Rahul CS 85 88 A
Sorted by Marks:
Name Department Marks Attendance Grade
1 Diya IT 92 90 A
2 Rahul CS 85 88 A
0 Aman CS 78 85 B
4 Arjun IT 74 80 B
3 Sneha ECE 67 75 C

13. Using the above dataframe answer the following questions.
a. Create a new column named “Total” by adding marks and bonus.
b. Find average marks department-wise.
c. Count the number of students in each grade category.
d. Calculate: mean, median,standard deviation of student marks.
e. Display students having maximum attendance.
f. Increase marks of all students by 5.

Input:

```
import pandas as pd
data = {
   "Student": ["A", "B", "C", "D", "E", "F", "G", "H"],
   "Department": ["CSE", "ECE", "CSE", "EEE", "ECE", "CSE", "EEE", "ECE"],
   "Marks": [75, 82, 68, 90, 76, 88, 65, 79],
   "Bonus": [5, 3, 7, 2, 4, 5, 6, 3],
   "Grade": ["B", "A", "C", "A", "B", "A", "C", "B"],
   "Attendance": [85, 92, 78, 95, 88, 95, 75, 90]
}
df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

df["Total"] = df["Marks"] + df["Bonus"]

print("\nDataFrame after adding Total:")
print(df)

average_marks = df.groupby("Department")["Marks"].mean()

print("\nAverage Marks Department-wise:")
print(average_marks)

grade_count = df["Grade"].value_counts()

print("\nNumber of Students in Each Grade:")
print(grade_count)

mean_marks = df["Marks"].mean()
median_marks = df["Marks"].median()
standard_deviation = df["Marks"].std()

print("\nStatistics of Student Marks:")
print("Mean =", mean_marks)
print("Median =", median_marks)
```

```
print("Standard Deviation =", standard_deviation)

maximum_attendance = df["Attendance"].max()

print("\nMaximum Attendance =", maximum_attendance)

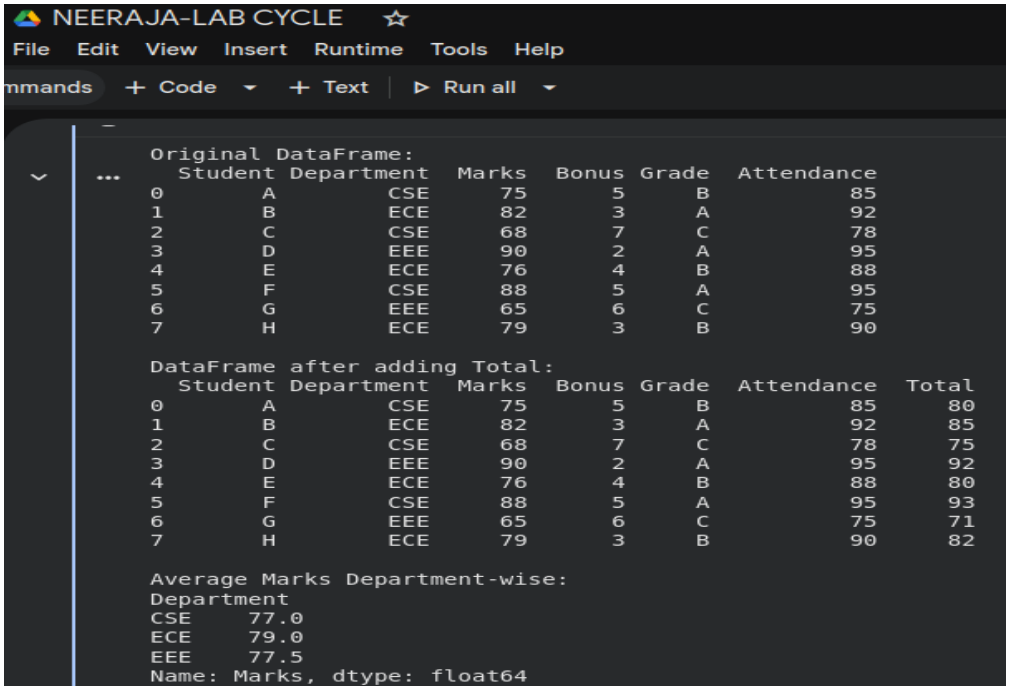
students_max_attendance = df[
    df["Attendance"] == maximum_attendance
]

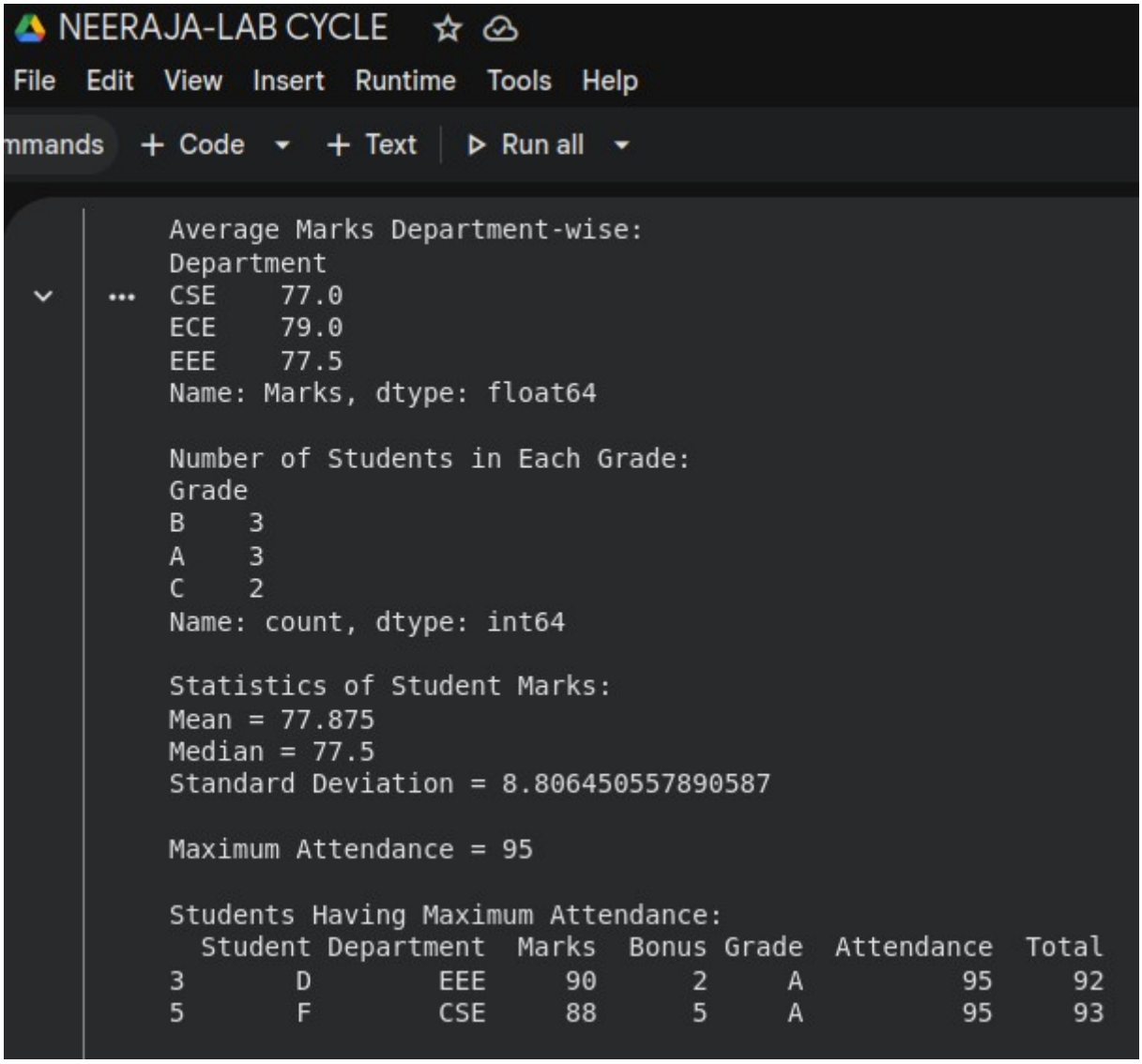
print("\nStudents Having Maximum Attendance:")
print(students_max_attendance)

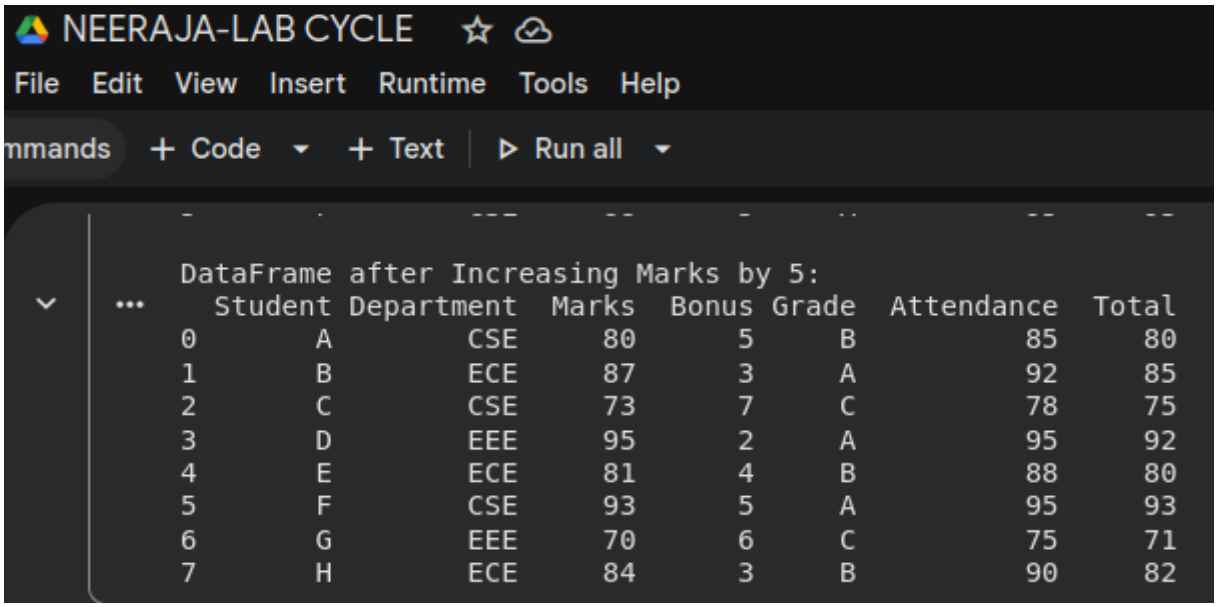

df["Marks"] = df["Marks"] + 5

print("\nDataFrame after Increasing Marks by 5:")
print(df)
```

Output:




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File Edit View Insert Runtime Tools Help
mmands + Code + Text Run all
Average Marks Department-wise:
Department
CSE 77.0
ECE 79.0
EEE 77.5
Name: Marks, dtype: float64
Number of Students in Each Grade:
Grade
B 3
A 3
C 2
Name: count, dtype: int64
Statistics of Student Marks:
Mean = 77.875
Median = 77.5
Standard Deviation = 8.806450557890587
Maximum Attendance = 95
Students Having Maximum Attendance:
Student Department Marks Bonus Grade Attendance Total
3 D EEE 90 2 A 95 92
5 F CSE 88 5 A 95 93


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File Edit View Insert Runtime Tools Help
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DataFrame after Increasing Marks by 5:
Student Department Marks Bonus Grade Attendance Total
0 A CSE 80 5 B 85 80
1 B ECE 87 3 A 92 85
2 C CSE 73 7 C 78 75
3 D EEE 95 2 A 95 92
4 E ECE 81 4 B 88 80
5 F CSE 93 5 A 95 93
6 G EEE 70 6 C 75 71
7 H ECE 84 3 B 90 82

14. Write a Python program using Pandas to clean and preprocess the following student dataset.
a. Perform the following operations:
i. Create a DataFrame containing student details.
ii. Identify missing values in the dataset.
iii. Replace missing values in the Marks column using the mean value.
iv. Remove duplicate rows from the dataset.
v. Convert student names into uppercase.
vi. Rename the column Marks to Score.
vii. Display the cleaned dataset.

Program:

```
import pandas as pd

data = {
    "Name": ["Anu", "Rahul", "Meera", "Arjun", "Anu", "Vijay"],
    "Age": [20, 21, 20, 22, 20, 21],
    "Marks": [85, 90, None, 75, 85, None],
    "Department": ["CSE", "ECE", "CSE", "EEE", "CSE", "ECE"]
}

df = pd.DataFrame(data)

print("Original Dataset:")
print(df)

print("\nMissing Values:")
print(df.isnull())

print("\nNumber of Missing Values in Each Column:")
print(df.isnull().sum())
mean_marks = df["Marks"].mean()

print("\nMean of Marks:", mean_marks)

df["Marks"] = df["Marks"].fillna(mean_marks)

print("\nDataset after replacing missing Marks:")
print(df)

df = df.drop_duplicates()

print("\nDataset after removing duplicate rows:")
print(df)
```

```
df["Name"] = df["Name"].str.upper()

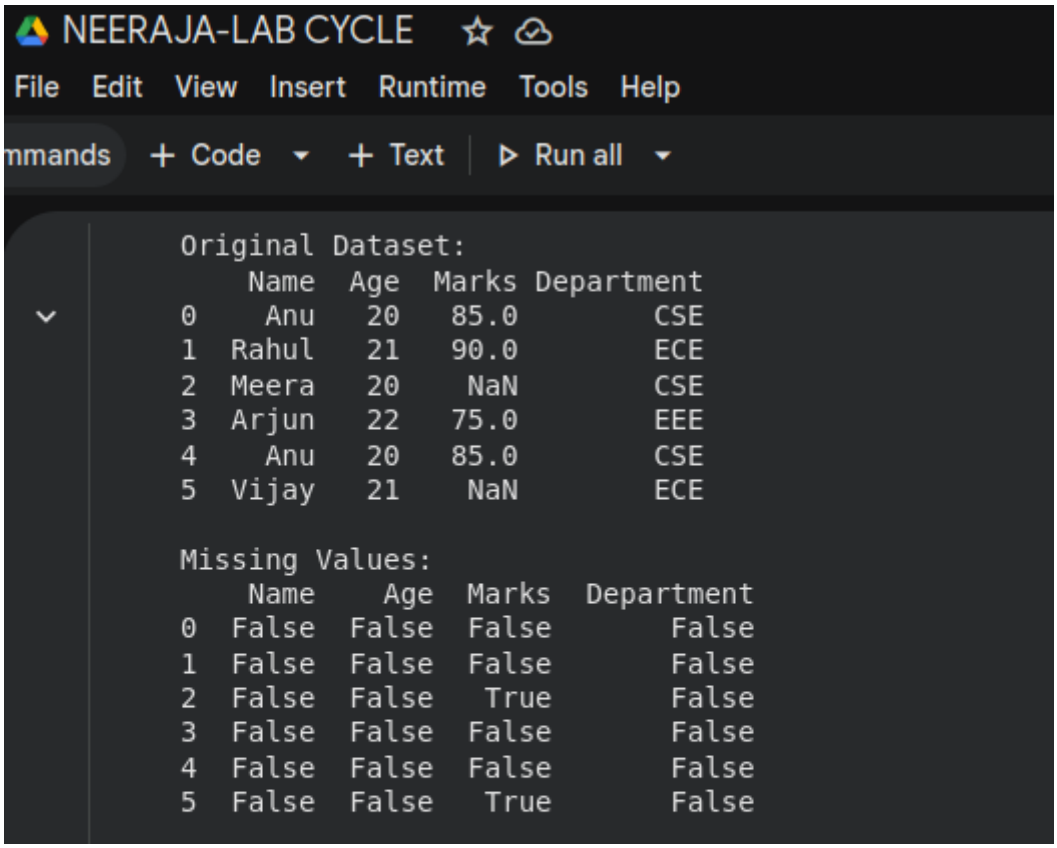
print("\nDataset after converting names to uppercase:")
print(df)

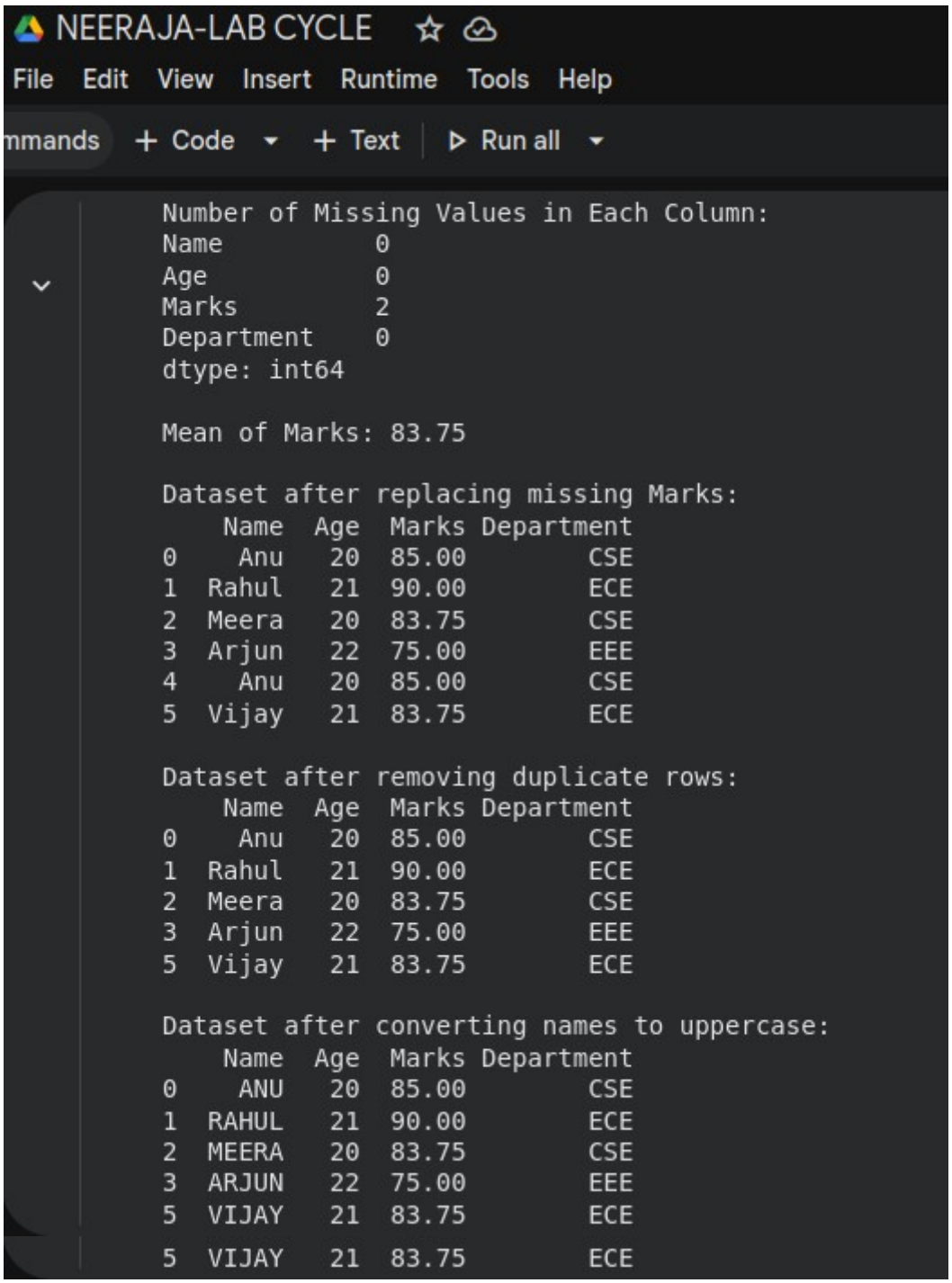
df = df.rename(columns={"Marks": "Score"})

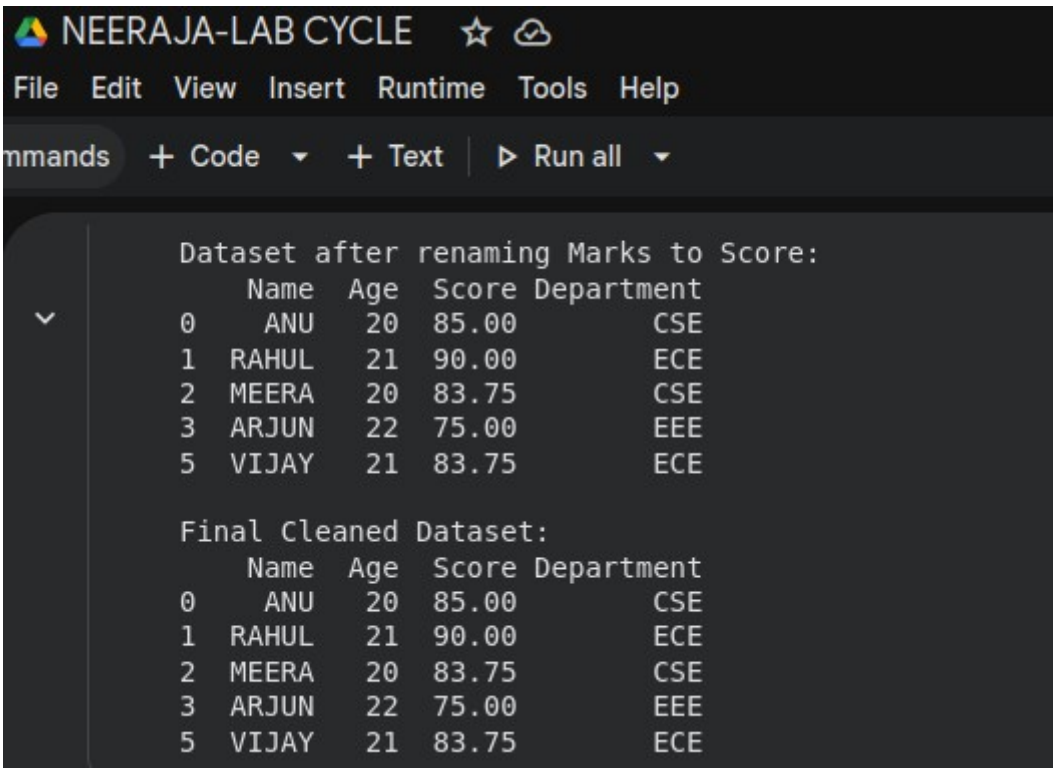
print("\nDataset after renaming Marks to Score:")
print(df)

print("\nFinal Cleaned Dataset:")
print(df)
```

## Output:




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File Edit View Insert Runtime Tools Help
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Number of Missing Values in Each Column:
Name 0
Age 0
Marks 2
Department 0
dtype: int64
Mean of Marks: 83.75
Dataset after replacing missing Marks:
Name Age Marks Department
0 Anu 20 85.00 CSE
1 Rahul 21 90.00 ECE
2 Meera 20 83.75 CSE
3 Arjun 22 75.00 EEE
4 Anu 20 85.00 CSE
5 Vijay 21 83.75 ECE
Dataset after removing duplicate rows:
Name Age Marks Department
0 Anu 20 85.00 CSE
1 Rahul 21 90.00 ECE
2 Meera 20 83.75 CSE
3 Arjun 22 75.00 EEE
5 Vijay 21 83.75 ECE
Dataset after converting names to uppercase:
Name Age Marks Department
0 ANU 20 85.00 CSE
1 RAHUL 21 90.00 ECE
2 MEERA 20 83.75 CSE
3 ARJUN 22 75.00 EEE
5 VIJAY 21 83.75 ECE
5 VIJAY 21 83.75 ECE


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Dataset after renaming Marks to Score:
Name Age Score Department
0 ANU 20 85.00 CSE
1 RAHUL 21 90.00 ECE
2 MEERA 20 83.75 CSE
3 ARJUN 22 75.00 EEE
5 VIJAY 21 83.75 ECE
Final Cleaned Dataset:
Name Age Score Department
0 ANU 20 85.00 CSE
1 RAHUL 21 90.00 ECE
2 MEERA 20 83.75 CSE
3 ARJUN 22 75.00 EEE
5 VIJAY 21 83.75 ECE

15.Plot simple mathematical functions.[y=3x^2+4x]

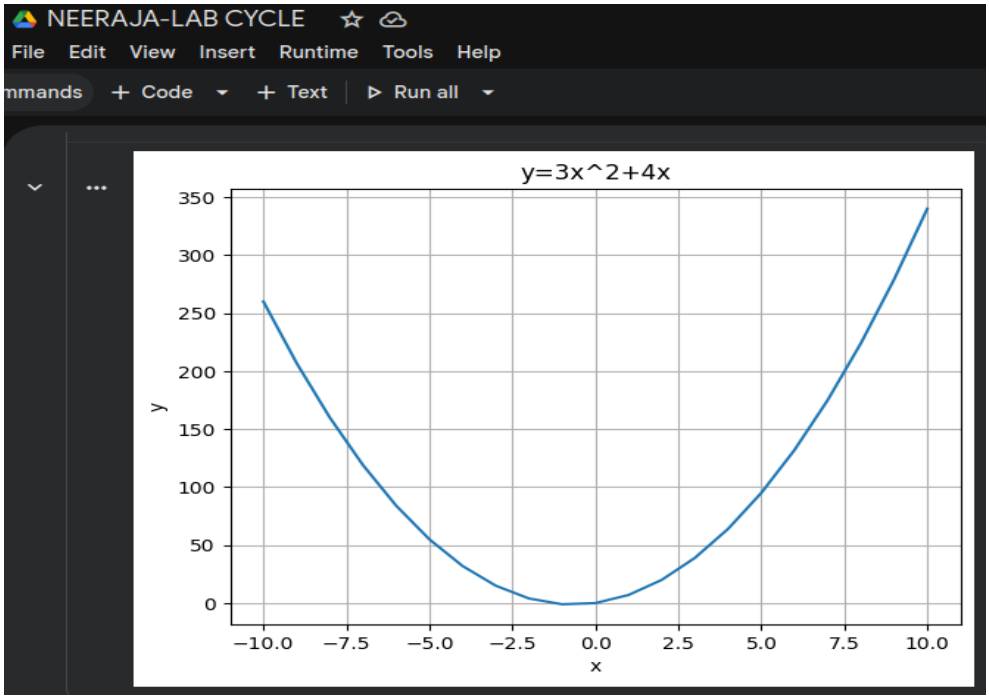
Program:

```
import numpy as np
import matplotlib.pyplot as plt

x=np.arange(-10,11)
y=3*x**2+4*x

plt.plot(x,y)
plt.xlabel("x")
plt.ylabel("y")
plt.title("y=3x^2+4x")
plt.grid()
plt.show()
```

Output:



## 16. Create simple login form using streamlit

## Program:

```
%%writefile form-gui.py
import streamlit as st

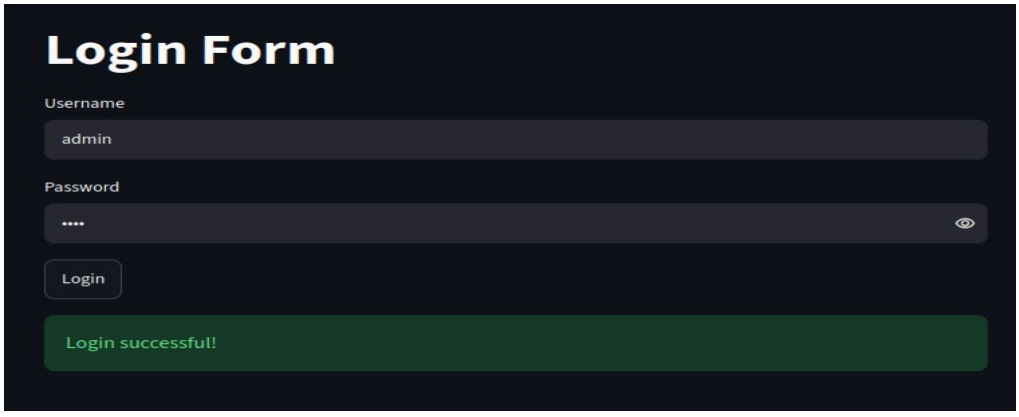
st.title("Login Form")

username = st.text_input("Username")
password = st.text_input("Password", type="password")

if st.button("Login"):

    if username == "admin" and password == "1234":
        st.success("Login successful!")
    else:
        st.error("Invalid username or password")
```

## Output:



## 17. Create student registration form using streamlit

## Program:

```
%%writefile resg-gui.py
import streamlit as st

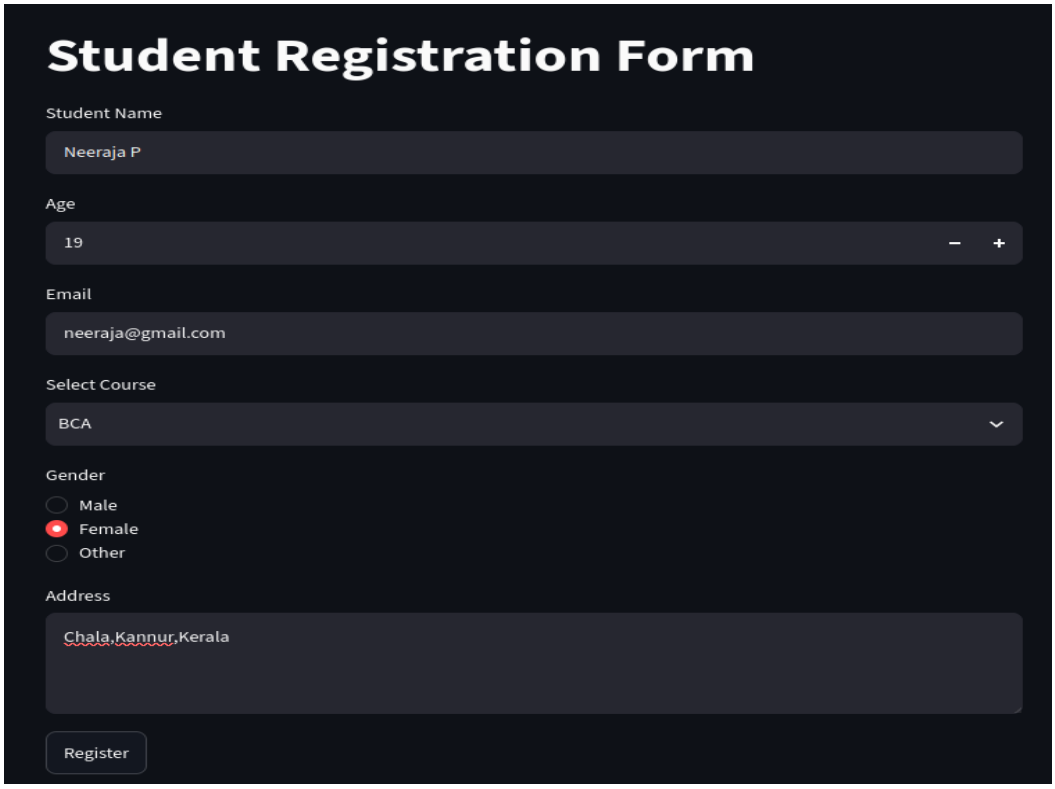
st.title("Student Registration Form")

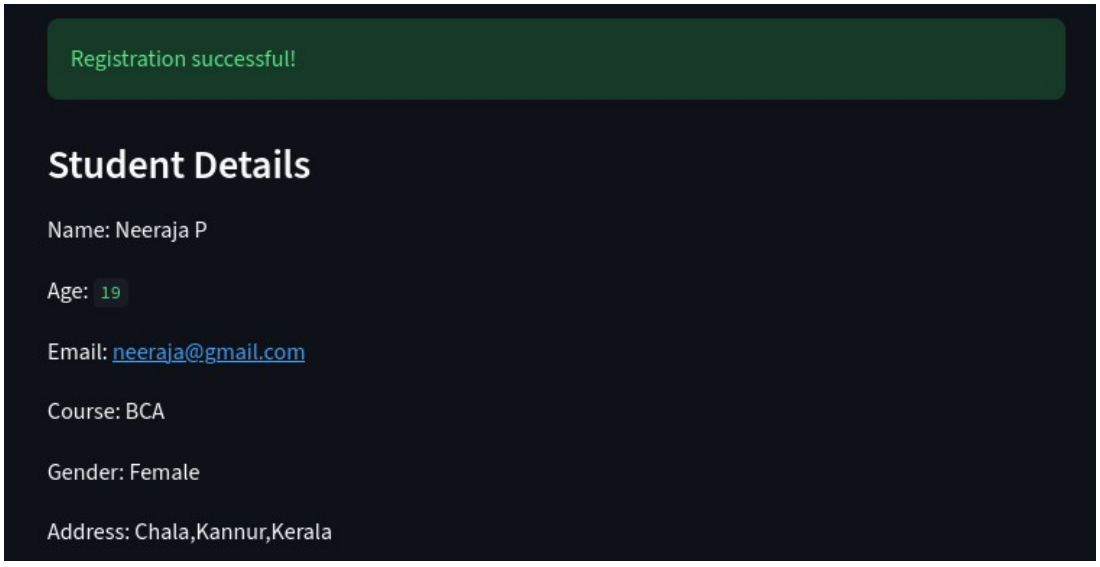
name = st.text_input("Student Name")
age = st.number_input("Age", min_value=1, max_value=100)
email = st.text_input("Email")
course = st.selectbox(
    "Select Course",
    ["BCA", "BSc Computer Science", "BCom", "BA"]
)
gender = st.radio("Gender", ["Male", "Female", "Other"])
address = st.text_area("Address")

if st.button("Register"):
    if name and email and address:
        st.success("Registration successful!")

        st.write("### Student Details")
        st.write("Name:", name)
        st.write("Age:", age)
        st.write("Email:", email)
        st.write("Course:", course)
        st.write("Gender:", gender)
        st.write("Address:", address)
    else:
        st.warning("Please fill in all required fields.")
```

## Output:


Student Registration Form
Student Name
Neeraja P
Age
19
Email
neeraja@gmail.com
Select Course
BCA
Gender
Male
Female
Other
Address
Chala,Kannur,Kerala
Register


Registration successful!
Student Details
Name: Neeraja P
Age: 19
Email: neeraja@gmail.com
Course: BCA
Gender: Female
Address: Chala,Kannur,Kerala